DOC 4 · Visualization System

Project: Public Pulse / Poll Platform Version: v1 Status: Planning Freeze (Implementation Open)

Purpose: Define how results should be transformed into experiences.

This document does NOT define:

- exact UI
- component library
- layouts
- colors
- final interactions

Those belong to design and implementation.

This document defines:

- visualization philosophy
- selection rules
- approved families
- interaction expectations
- reference interpretation

---

1. Visualization Philosophy

Visualization is not reporting.

Visualization is reward.

Users should feel:

“I want to see what happened.”

not

“I opened analytics.”

Every visualization should optimize:

1. Result Interestingness

2. Participation Reward

3. Exploration

4. Discussion

5. Shareability

Secondary: Data utility

---

2. Core Principles

V1

Visualization chosen per QUESTION.

Never: Category → fixed visualization

Always:

Question ↓

Answer structure ↓

Data structure ↓

Visualization

V2

One Question = One Experience

Question

Answer

Result

Exploration

exist together.

Different states. Same experience.

V3

Primary + Secondary

Every question should attempt:

1 Primary visualization

+

0–2 Secondary visualizations

Primary: instant understanding

Secondary: curiosity

Avoid: visual overload

V4

Reference Images

Reference images are:

Strong constraints WITH adaptation.

Allowed:

- recomposition
- simplification
- combination
- evolution

Not allowed: copying literally.

If chosen references fail:

Design agent may replace.

Replacement must explain:

Why What improved What was preserved

---

3. Visual Language

Target:

Modern

Smooth

Interactive

Mobile-first

Medium information density

Editorial

Avoid:

Corporate dashboard

Dense BI

Text-first

Static charts

Box overload

---

4. Motion System

Goal: Make state transitions understandable.

Default:

Immediate acknowledgement

↓

300–700ms motion

↓

settle

Heavy: <900ms

Never: cinematic delays

Motion verbs:

Reveal

Expand

Highlight

Flow

Cluster

Settle

Motion should explain.

Not decorate.

---

5. Result Visibility

Visibility is QUESTION dependent.

Examples:

Low sample: placeholder

Enough sample: show

Special questions: unlock

No universal rule.

Thresholds: implementation decision.

---

6. Result Refresh

Refresh events:

User answered

Periodic refresh

Manual refresh

Builder decides actual cadence.

Recommendation: avoid live twitching.

---

7. Data Transparency

Default:

Votes

+

Sample size

Optional:

Raw

Weighted

Methodology

Sample size visible.

Never dominant.

---

8. Personal Layer

Mandatory.

Every question should attempt:

Your Answer

+

One Insight

Examples:

You matched X%

You differed

Your city differs

Minority

Movement

Insight rules:

Template driven.

Not AI generated.

---

9. Visualization Families

Status definitions:

MVP

Conditional

Future

S · Split / Opinion

Purpose: Fast comparisons.

Approved:

S1 Split Cards (MVP)

S2 Radial Split (MVP)

S3 Liquid Fill (MVP)

S4 Cluster Split (Conditional)

Rules:

Simple first.

Motion matters.

Avoid: analytics feel.

References: DV reference board.

G · Geographic

Purpose: Spatial understanding.

Approved:

G1 Gradient Heat Map (MVP)

G2 Bubble Map (MVP)

G3 Pattern Distribution (Conditional)

Rules:

Use ONLY when geography matters.

Never force map.

R · Ranking

Purpose: Competition.

Approved:

R1 Tier Board

R2 Dynamic Leaderboard

R3 Podium

R4 Packed Ranking (Conditional)

Rules:

Prefer labels.

Avoid: S/A/B.

T · Tournament

Purpose: Long-running engagement.

Approved:

T1 Bracket

T2 Survivor Tree (Conditional)

Rules:

Product mode.

Not question type.

F · Flow

Purpose: Movement.

Approved:

F1 Sankey (Future)

Rules:

Needs enough data.

N · Network

Purpose: Relationships.

Approved:

N1 Network

N2 Cluster Graph

(Conditional)

P · Proportion

Purpose: Distribution.

Approved:

P1 Treemap

P2 Circle Packing

(Conditional)

I · Insight

Purpose: Personal meaning.

Approved:

Insight Cards

Comparison Cards

(MVP)

---

10. Selection Framework

Evaluate every DV:

Participation Value

Result Interestingness

Visualization Fitness

Data Utility

Interaction Quality

Shareability

Mobile Experience

Complexity

Reject if:

Hard to understand

Only pretty

Requires explanation

Too dashboard-like

---

11. Expansion Rules

Question opens

↓

Primary DV

↓

Expand

↓

More DVs

Recommended: 2–3 visualizations maximum.

Question decides.

Not global.

---

12. Builder Challenge Checklist

Builder should ask:

Can simpler work?

Can motion help?

Can result become addictive?

Can this become shareable?

Can interaction reduce effort?

Can another DV explain better?

Challenge all selected references.

Keep intent.

---

13. Design Agent Instructions

Interpret.

Do not inherit.

Design owns:

layout

spacing

transitions

consistency

component language

Respect:

information density

visual hierarchy

interaction expectations

Final Principle

Good visualization explains.

Great visualization invites participation.